

PRASAD CHOPADE

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Education

Sardar Patel Institute of Technology (SPIT), Mumbai

2022 - 2026

Bachelor of Technology (B.Tech), Computer Engineering

CGPA: 8.96

Relevant Coursework: Software Engineering, Artificial Intelligence and Machine Learning, Database Management System, Design and Analysis of Algorithms, Operating Systems, Data Structures, Deep Learning, Data Analytics

Experience

Software Developer Intern - Adivid Technologies

July 2025 – Dec 2025

Project: Development of 'The Hub', an Integrated Course and Management System for ICEP Europe

- Delivered single sign-on course access for 3,000+ student accounts by implementing token-based authentication and a Moodle LMS integration that synced enrollment and progress data in real time.
- Built 4 REST API-driven admin modules - user, course, enrollment, and learner record management in Node.js, Express, and PostgreSQL, replacing spreadsheet DB workflow the ICEP admin team used to manage learner records.
- Implemented role based authorization and secure API access controls to safeguard student records and administrative operations across the learning management platform.

Research Intern - Indian Institute of Technology (IIT), Guwahati

Jan 2025 – June 2025

Project: Intrusion Detection System for 6G-Enabled UAV-V2X Networks

- Designed and implemented a real-time Intrusion Detection System (IDS) for 6G-enabled UAV and vehicular networks using NS-3 and PyTorch, integrating it seamlessly with the ONNX runtime for live in-loop inference.
- Engineered a large-scale dataset from simulated UAV-V2X environments, performing data cleaning, feature engineering, and statistical validation for ML readiness and predictive modeling.
- Developed system analytics and performance metrics to assess resilience against five cyberattack vectors, achieving 91.5% detection recall and actionable insights on network vulnerabilities.

Projects

Federated Learning for Medical Image Diagnosis | PyTorch, ResNet-50, Federated Learning

- Built an end-to-end federated learning pipeline in PyTorch training a ResNet-50 classifier from random initialization across 144,660 chest X-rays from three public datasets (NIH ChestX-ray14, RSNA, Pediatric), with naturally non-IID adult-vs-pediatric distributions and a unified 3-class label space.
- Achieved 80% average accuracy across all three hospitals using importance-weighted aggregation, outperforming both isolated local training (76.7%) and standard FedAvg (77.2%).
- Designed a round-wise client scheduling strategy that raised one hospital's accuracy from 77.7% to 83%, isolating the impact of participation scheduling from aggregation weighting alone.

Saarthi - Learning Platform | React, TailwindCSS, Node.js, Python, AWS, Docker, Youtube API

- Developed a personalized AI learning platform integrating roadmap generation, adaptive content delivery, and AI-assisted note taking, improving learner engagement through data driven personalization.
- Built and deployed an RAG based chatbot that analyzed 100+ hours of learning videos weekly, generating resources with high relevancy and supporting intelligent content retrieval via scalable, rate limited architecture.

AI Agent Workflow Builder | Next.js, TypeScript, Hasura, PostgreSQL, GraphQL, Docker

- Built and deployed a multi-tenant AI workflow engine on Next.js and Hasura supporting 6 step types and 4 trigger types, with runs pausing at human approval gates and streaming per-step status live over GraphQL subscriptions.
- Secured the platform so one company's data is completely unreachable by another, with access rules enforced in the database layer rather than in application code; wrote an automated attack suite that tries to read, modify and approve another company's records using their exact record IDs - 37 checks, every unauthorized attempt blocked.
- Prevented simultaneous requests from pushing an organization past its monthly usage limit, and made repeated or duplicated requests safe so a workflow never runs twice by accident - verified by 112 automated tests covering the database, access rules, failure retries and the full user journey.

Publications

- Systematic Evaluation of CNN-Transformer Hybrid Architectures for Multi-Class Waste Classification (IEEE IntelliSecAI 2025)
- Enhanced Federated Learning for Medical Image Diagnosis using Reinforcement Learning (ISED 2026 - NIT Warangal)
- NS-3/mmWave-Based Simulation Framework for Real-Time Anomaly Detection in UAV Assisted 6G Networks (COMSNETS 2026)

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C, SQL, HTML/CSS

Frameworks Libraries: React, Next.js, Node.js, Express, Flask, PyTorch, Pandas, NumPy, OpenCV

Data and Infrastructure: PostgreSQL, GraphQL, Hasura, Docker, AWS, Git, MongoDB

Extracurricular

- Operations Head, TechRace (Oculus, SPIT): Led the planning and on-ground execution of a city wide, app based treasure hunt engaging 500+ participants, coordinating logistics, resources, and team operations.
- Secured 2nd position (Team ByteMe) in StatHack, a national level data science hackathon organized by IIT Goa.